

Epochs and Monitoring Training

Deep Learning

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Contents of This Video

In this video, we will cover:

- What is an epoch?
- Reading training curves
- Diagnosing underfitting vs. overfitting
- When to stop training
- Good vs. bad training curves

What is an Epoch?

One epoch = one complete pass through the training data

Example:

- Training set: 10,000 samples
- Batch size: 100
- Batches per epoch: $10,000 / 100 = 100$
- 1 epoch = 100 weight updates

Typical training: 10-100+ epochs

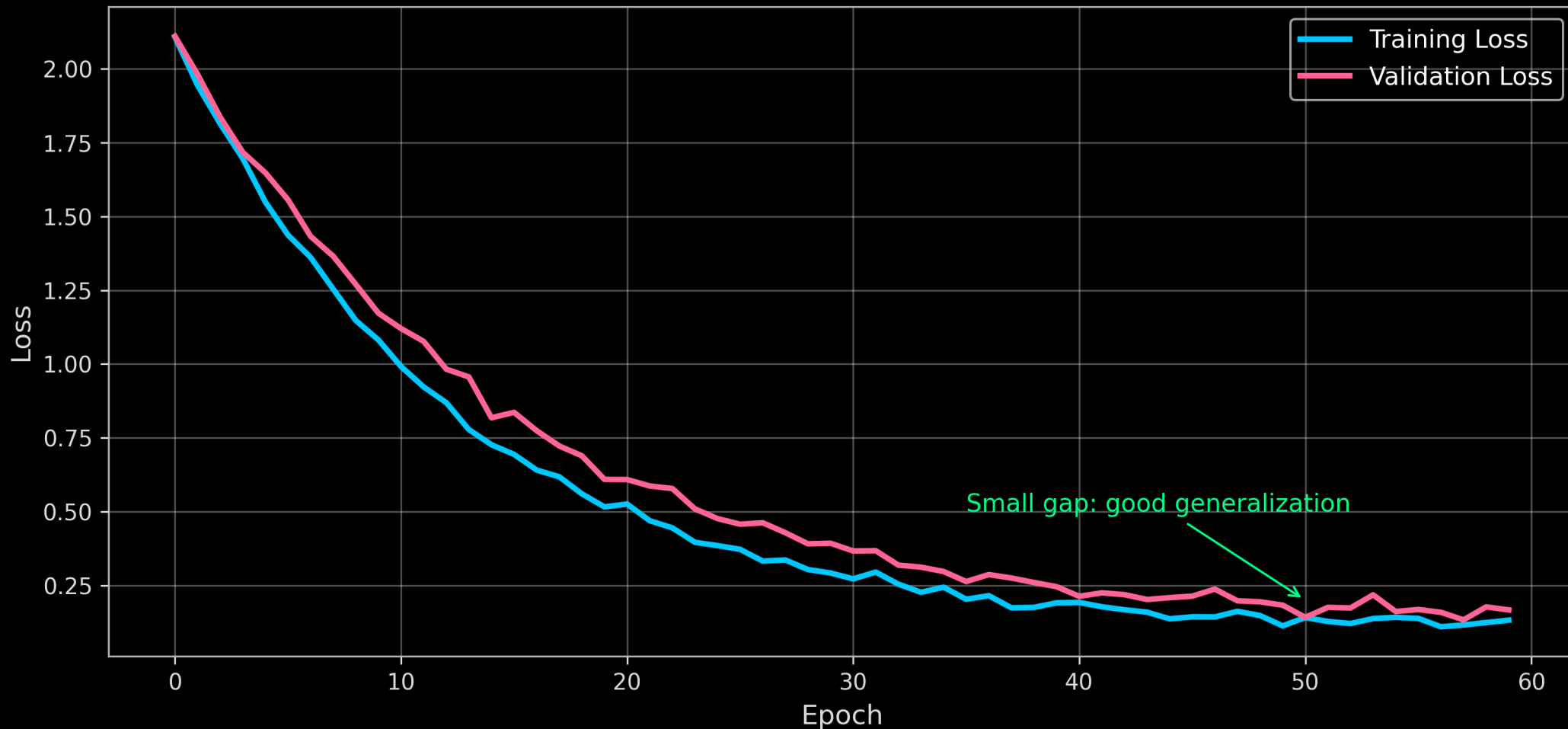
Monitoring: Train and Validation Loss



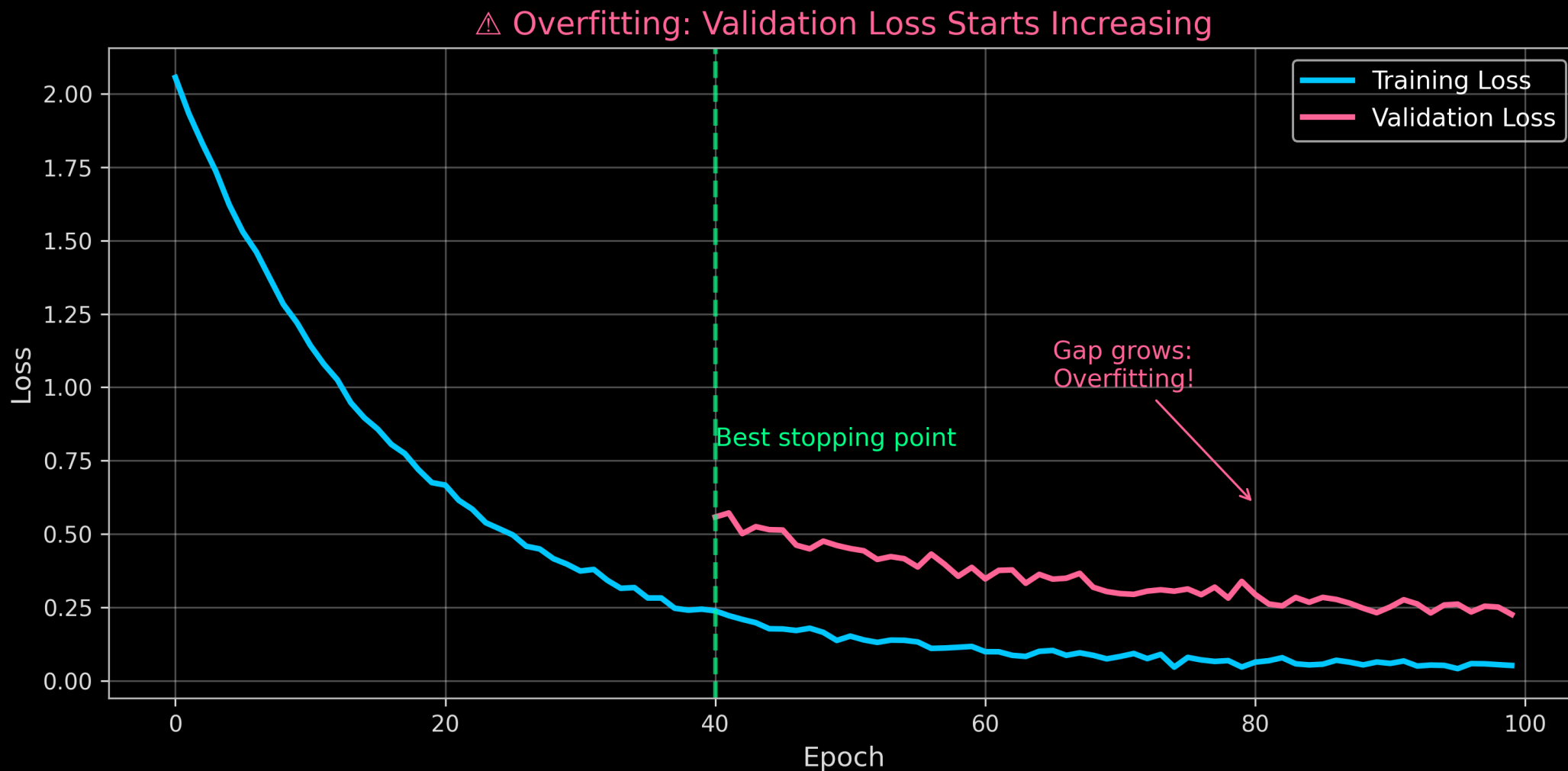
Always track both curves—they tell different stories!

Good Training: Both Losses Decrease

✓ Good Training: Both Curves Decrease, Small Gap

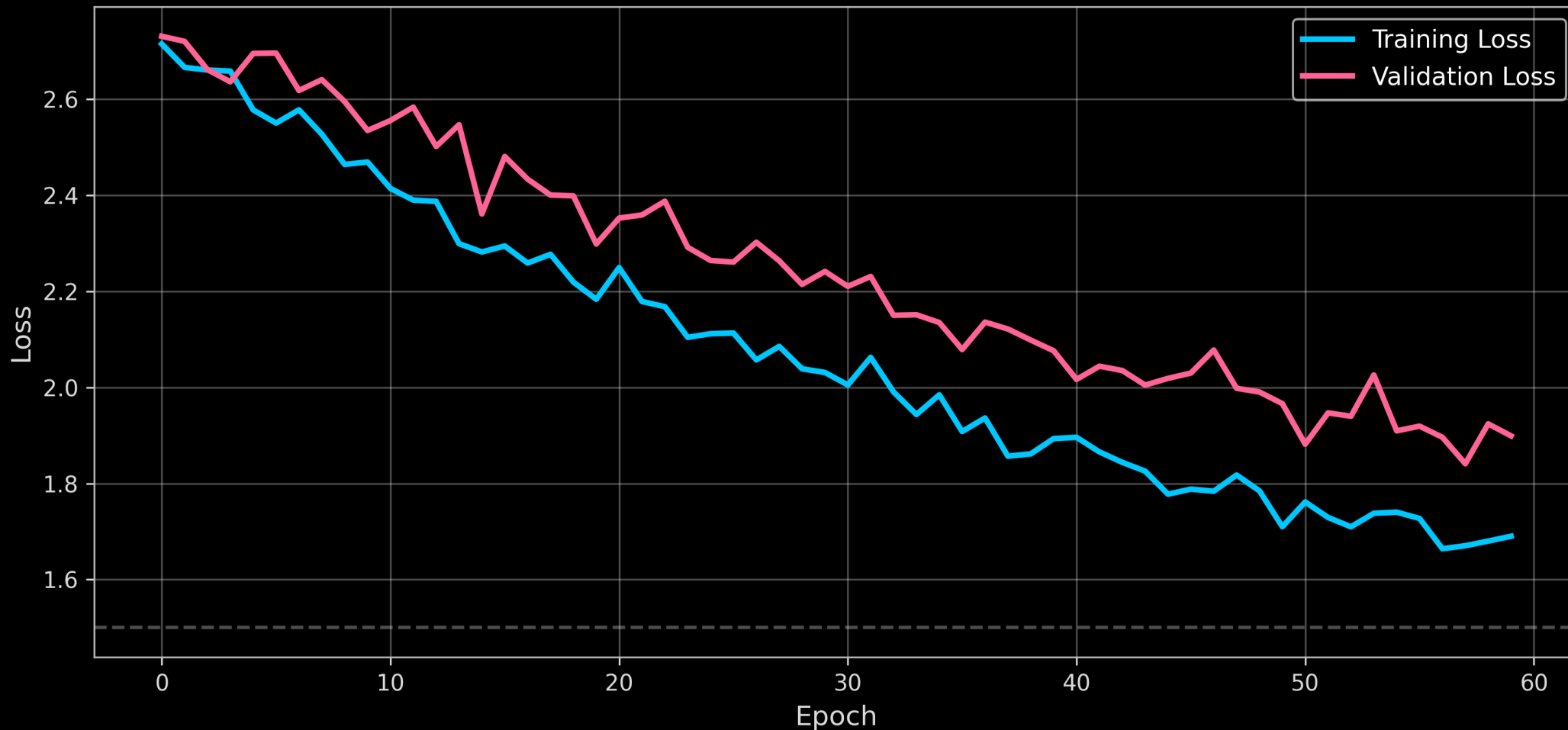


Overfitting: Validation Loss Increases



Underfitting: Both Losses High

⚠ Underfitting: Both Losses Remain High



Model isn't powerful enough to capture the patterns

Diagnostic Summary

Observation	Diagnosis	Solution
Both losses high	Underfitting	Bigger model, longer training
Train low, val high	Overfitting	Regularization, early stopping
Both losses decrease	Good!	Keep training
Loss oscillates	LR too high	Reduce learning rate
Loss barely changes	LR too low	Increase learning rate

When to Stop Training

Stopping criteria:

1. **Validation loss stops improving** (best criterion)
2. Maximum epochs reached
3. Training loss below threshold
4. Time/compute budget exhausted

Early stopping = stop when validation loss hasn't improved for N epochs

- Typically $N = 3-10$ (“patience”)
- Save the best model (lowest validation loss)

What We've Covered

In this video, we've learned:

- One epoch = one pass through all training data
- Monitor both training and validation loss
- Good training: both decrease with small gap
- Overfitting: validation loss increases while training decreases
- Underfitting: both losses remain high
- Stop when validation loss stops improving